

# Omar AbdElzaher

Senior Unity Developer / Systems Architect · Multiplayer · DOTS/ECS · VR/XR

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Egypt (UTC+2) · Open to remote, hybrid & on-site roles internationally · Available to relocate

## SUMMARY

Senior Unity developer with 10+ years shipping multiplayer, VR/XR, and cross-platform titles from prototype to live release. Architected *Highstreet: Calamity*, a co-op VR roguelike on Meta Quest, on a DOTS-based, server-authoritative backend powered by Amazon GameLift. Specialize in scalable networked systems, ECS/DOTS architecture, and leading teams; active open-source contributor (200+ stars, 40+ repos), including a popular ComfyUI AI-tooling plugin.

## TECHNICAL SKILLS

**Core:** Unity 6 (DOTS/ECS, MonoBehaviour) · Multiplayer Systems · VR/XR

**Languages:** C# · JavaScript · C/C++ · Python

**Networking:** Netcode for Entities · Netcode for GameObjects (NGO) · Unity Transport (UTP) · Amazon GameLift · Matchmaking · SmartFox

**Architecture:** DOTS/ECS · Data-Driven Design · ScriptableObjects · Behaviour Trees · Procedural Generation · Editor Tooling · Design Patterns

**AI & Generative:** LLM APIs (OpenAI, Claude) · Image/Video/Audio Generation (Stable Diffusion, ComfyUI) · Prompt Engineering · AI Agents & Tool Calling · AI-Assisted Development

**Performance:** Unity Profiler · Object Pooling · Optimization

**Platforms:** VR/XR · Mobile · Web/WebGL · PC

**Tools:** Git · GitHub Actions (CI/CD) · Firebase · Agile · Node.js

## EXPERIENCE

### Independent Game Developer · Self-employed

2026 – Present · Remote

- Co-developing an unreleased PC shop-management simulation in Unity (C#) within a two-person indie team, targeting release in ~2 months.
- Implemented data-driven, ScriptableObject-based systems spanning inventory, economy, loot generation, and customer AI, plus a modular interaction system and NPC behaviour-graph AI.
- Contribute across gameplay systems, UI, tooling, and content pipeline end-to-end within the team.
- Leverage AI-assisted tooling (LLM/agent workflows and generative media APIs) to accelerate prototyping and content creation.

### Game Systems Architect · Highstreet

Jan 2024 – Mar 2026 · Remote

- Architected *Highstreet: Calamity* (Meta Quest), a co-op VR roguelike with real-time multiplayer, procedural gameplay, and a social hub — live on the Meta Store.
- Defined a DOTS-first, data-driven architecture (visual system diagrams + Excel config) enabling scalable simulation of 1,000–10,000 entities at 20 Hz with clear system ownership.
- Implemented server-authoritative systems (combat, AI, abilities, inventory) with Netcode for Entities, supporting up to 100 concurrent players at sub-50 ms latency.
- Built a custom ECS-compatible Behaviour Tree with a visual editor, adopted as the team's primary AI tool.
- Migrated matchmaking/backend from Unity Multiplay to Amazon GameLift with zero downtime; mentored 3–5 engineers into independent DOTS system ownership.

### Senior Game Developer · Highstreet

2022 – Dec 2023 · Remote

- Shipped *Highstreet Campus*, a WebGL social MMO hub; designed and implemented a custom networking layer supporting up to 100 concurrent players.
- Designed a distributed server architecture (authoritative core + regional workers) that scaled dynamically with player presence while holding sub-50 ms p95 latency.
- Built a unified inventory system integrating NFTs, backend services, and in-game assets via adapter-based design.

- Integrated separate VR and WebGL experiences into one ecosystem with seamless player transitions.

#### Game Development Team Lead · Zinad IT / EntrenchTech

Feb 2020 – 2022 · Egypt

- Led 4+ developers delivering 10+ serious cybersecurity-awareness games across VR, mobile, WebGL, and Kinect.
- Architected a unified multi-platform codebase, replacing fragmented builds with a mobile-first system that cut average delivery time from 6 months to 1–2 months per title.
- Delivered enterprise VR and interactive training experiences for clients under NDA.
- Established coding standards and mentored junior developers.

#### Game Track Supervisor & Teaching Assistant · Information Technology Institute (ITI)

Aug 2018 – Feb 2020 · Egypt

- Designed and delivered a 9-month Unity curriculum covering C/C++, design patterns, VR, optimization, and Git.
- Mentored cohorts of 20+ students per intake.

#### Game Developer · A15 / El3ab.com

May 2015 – Oct 2017 · Egypt

- Rebuilt and shipped *Estimation Kings*, migrating it from Flash to Unity for Web, Android, and iOS.
- Developed a real-time multiplayer system with global matchmaking using SmartFox Server.
- Implemented server-authoritative gameplay architecture and MVC-based game systems.

### OPEN-SOURCE PROJECTS

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#### DOTS Behaviour Tree (co-developed with M-Hamdy23)

Unity / C# / DOTS · [github.com/M-Hamdy23/DotsBehaviourTree](https://github.com/M-Hamdy23/DotsBehaviourTree)

ECS-compatible behaviour tree system with visual editor and runtime debugging tools.

#### ScriptableObject Framework (SO\_System)

Unity / C# · [github.com/omar92/SO\\_System](https://github.com/omar92/SO_System)

ScriptableObject-based framework for data-driven, event-driven, loosely coupled systems with integrated variables, events, UI binding, and editor tools.

#### Optimized Unity Scroll List

Unity / C# · [github.com/omar92/Optimized-unity-scroll-list](https://github.com/omar92/Optimized-unity-scroll-list)

Virtualized, object-pooled scroll list rendering unlimited items with minimal draw calls and lifecycle callbacks.

#### ComfyUI Quality-of-Life Suite ★ 170 · 25 forks

Python · AI tooling · [github.com/omar92/ComfyUI-QualityOfLifeSuit\\_Omar92](https://github.com/omar92/ComfyUI-QualityOfLifeSuit_Omar92)

Popular ComfyUI plugin adding ChatGPT/DALL-E nodes, AI prompt enhancer, and text-to-image tools.

### EDUCATION

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#### Professional Diploma in Game Development — Information Technology Institute (ITI)

2017 – 2018

Unity3D · VR · C/C++ · Design Patterns · Optimization | Grade: Very Good | Graduation: Rhythm Attack VR

#### B.Sc. Computer Engineering — Misr Higher Institute for Engineering and Technology

2008 – 2013

Focus: Software Engineering · Computer Graphics · AI · Systems Architecture | Grade: Very Good

### ADDITIONAL

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**Languages:** Arabic (Native) · English (Fluent)

**Portfolio:** [omar92.github.io/portfolio](https://omar92.github.io/portfolio)